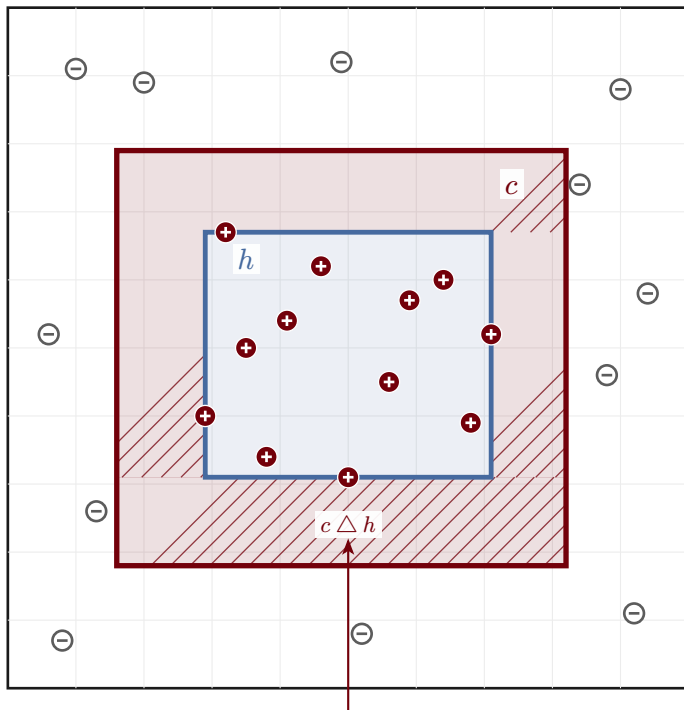


PAC learning: target c , hypothesis h , error region

$$\Pr_{S \sim D^m} [\text{err}(h) \leq \varepsilon] \geq 1 - \delta \quad \text{when} \quad m \geq \left(\frac{4}{\varepsilon}\right) \ln\left(\frac{4}{\delta}\right)$$



- target concept c
- hypothesis h (tightest fit)
- error region $c \triangle h$
- + positive (+) sample
- − negative (−) sample

Every positive draw lies in c , so $h \subseteq c$ and $c \triangle h = c \setminus h$. Generalization error is the probability mass of this frame.

$$\text{err}(h) = D(c \triangle h) \leq \varepsilon$$

instance space $\mathcal{X}$